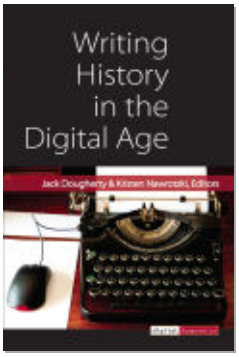
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Visualizations and Historical Arguments

可视化与历史论证

John Theibault

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The popular aphorism “A picture is worth a thousand words” is a relatively recent coinage, but the idea that images can be an effective complement to or substitute for written description, narrative, or analysis is probably as old as writing itself. In the European tradition, illuminated manuscripts and incunabula incorporated images, some of which conveyed messages related to the text and some of which were mere adornments. By the late sixteenth century, linkage of image and print reached a kind of apotheosis with the publication of emblem books, in which each page consisted of an image, a motto, and a pithy verse that jointly communicated moral precepts.1

“一图胜千言”这句流行格言其实出现得相对较晚，但图像可以有效补充甚至替代文字描述、叙事或分析这一观念，很可能与文字本身一样古老。在欧洲传统中，彩饰手抄本和早期印刷本中都包含图像，其中一些传达与文本相关的信息，另一些则只是装饰。到16世纪晚期，随着徽记书的出版，图像与印刷文字的结合达到了一种巅峰：每一页由一幅图像、一句格言和一段简短有力的诗句组成，三者共同传达道德训诫。1

Western historians have always made use of visualizations in this broad sense. Reproductions of pictures of the main biographical figures referenced in a book or of other objects that figured prominently in the narrative appear in many historical works. Though the connection of these illustrations to the arguments of the book were often implicit rather than explicit, the text sometimes drew direct attention to elements of the pictures, so that the reader’s understanding was enhanced by close attention to the images.

西方历史学家一直在这种广义意义上使用可视化。许多历史著作都会复制书中主要传记人物的肖像，或叙事中占有重要位置的其他对象。虽然这些插图与全书论证之间的联系往往是隐含的，而不是明确说出的，但正文有时也会直接提醒读者注意图像中的某些元素，使读者通过仔细观察图像而加深理解。

When the term visualization is used today, it usually refers to an image that is derived from processing information—often, but not always, statistical information—and that presents the information more efficiently than regular text could. Scholars quickly recognized the potential of computers to help process information and display the results in an easily interpreted format. David Staley has argued for a sharp distinction between visualizations as “the organization of meaningful information in two- or three-dimensional spatial form intended to further a systematic inquiry” and images as a “supplement or illustration to a written account.”2 Staley’s definition implies two distinct uses for visualizations in the digital age: as a means of quickly identifying patterns in large datasets during the research process, which can open new lines of research and test qualitative assumptions; and as a way to enhance the presentation of arguments, moving beyond what it is possible to display in two dimensions on paper. Visualizations created for the first use may or may not appear in visual form in the final product. This essay is primarily concerned with visualizations as historical arguments in the second sense: how do we deploy the visual capabilities of the computer to show what we wish to communicate? It is slightly more ecumenical in defining visualizations than Staley is, in that it sees as forms of visual argument all uses of visual information to communicate an argument or narrative beyond the meaning of the words in text. It argues that visualizations necessarily have a rhetorical dimension and that the principal challenge facing historians who wish to use visualizations in their work is to align the rhetoric with the audience’s ability to follow it. (See the images in the web version of this essay at http://WritingHistory.trincoll.edu.)

今天所说的“可视化”，通常是指通过处理信息——往往是统计信息，但并不总是如此——而生成的图像，并以比普通文字更高效的方式呈现这些信息。学者很快认识到，计算机能够帮助处理信息，并把结果以容易理解的形式展示出来。David Staley主张，应当把可视化——即“为了推进系统性探究，而将有意义的信息组织成二维或三维空间形式”——与作为“书面叙述的补充或图解”的图像严格区分开来。2 按照Staley的定义，数字时代的可视

化具有两种不同用途：一是在研究过程中快速识别大型数据集中的模式，由此开启新的研究路径并检验定性判断；二是强化论证的呈现方式，突破纸面二维展示的限制。为第一种用途制作的可视化，最终成果中未必会以视觉形式出现。本文主要关注第二种意义上的“作为历史论证的可视化”：我们如何利用计算机的视觉能力，把自己希望传达的内容展示出来？本文对可视化的定义比Staley稍为宽泛，把一切借助视觉信息、超越纯文字含义来传达论证或叙事的方式，都视为视觉论证。本文认为，可视化必然具有修辞维度；对于希望在研究中使用可视化的历史学家而言，最主要的挑战，是让这种视觉修辞与受众理解它的能力相匹配。（本文网页版中的图像见http://WritingHistory.trincoll.edu。）

The key dimensions of a visualization are the density and the transparency of its information. Density is the sheer amount of useful information the visualization conveys, and transparency is the ease with which that information can be understood by the reader. We have become so accustomed to the visual vocabulary of print books that we scarcely register the visual conventions on which almost all historical work relies, such as the footnote indicated by a small number or asterisk. By now, we are also perhaps so familiar with standard web-page layouts that we no longer notice most of the visual cues that indicate the site structure, especially the relation of one page to another achieved by hyperlink. But many historians still have a print mentality when it comes to information-dense graphics. When designing graphics, authors have to consider how much background information the reader brings to the visualization. The development of more complex visualizations has increased the gaps between expert and novice interpreters, which raises challenges for historians who seek the most effective visual approach.

可视化的两个关键维度是信息的密度和透明度。所谓密度，是可视化所传达的有用信息的数量；所谓透明度，是读者理解这些信息的容易程度。我们已经太习惯印刷书籍的视觉词汇，以至于几乎意识不到几乎所有历史著作都依赖的视觉惯例，例如用小号数字或星号标示脚注。如今，我们也可能已经十分熟悉标准网页布局，以至于不再留意那些提示网站结构的视觉线索，尤其是通过超链接形成的页面之间的关系。不过，在面对高信息密度的图形时，许多历史学家仍然保留着一种“印刷思维”。设计图形时，作者必须考虑读者在进入可视化之前已经具备多少背景知识。随着可视化变得更加复杂，专家读者与初学者之间的理解差距也在扩大，这给试图寻找最有效视觉表达方式的历史学家提出了挑战。

The problem of information density in visualizations is not a new one for historians. Even the most conventional nineteenth-century political histories made use of three important visualizations: maps, time lines, and dynastic charts. It is, after all, much easier and more informative to create a chart of lines of descent to the kings of France than it is to describe the lineage in paragraphs of “begats.” Each of these forms of visualization evolved a distinct visual vocabulary, with periods of experimentation and innovation producing visualization schemes that most modern historians now find completely transparent, with earlier visual dead ends now completely forgotten. Daniel Rosenberg and Anthony Grafton have recently shown how experimentation with designs of chronologies helped produce the modern streamlined edition of the time line and information-rich variants by the end of the eighteenth century.3

可视化中的信息密度问题，对历史学家来说并不是新问题。即使是最传统的19世纪政治史，也会使用三种重要的可视化形式：地图、时间线和王朝谱系图。毕竟，与其用一段又一段“某某生某某”的文字来描述法国国王的世系，不如直接画出一张谱系图，既容易理解，也能提供更多信息。这些可视化形式都逐渐形成了各自独特的视觉词汇；经过一段时期的试验与创新，最终产生了现代历史学家如今已经觉得完全直观透明的表达方案，而早期失败的视觉方案则几乎被彻底遗忘。Daniel Rosenberg和Anthony Grafton近来指出，对年代编排方式的反复试验，如何在18世纪末促成了现代简洁化时间线，以及信息量更丰富的时间线变体。3

The emergence of the social sciences in the nineteenth century and the ability to work with large datasets created demand for new ways of visualizing information, beyond maps, time lines, and genealogical charts. Processed numerical information was best expressed in tables, charts, and graphs. Mathematics, natural sciences, and social sciences that employed statistics were at the forefront of the development of charts and graphs. History was a consumer, not a designer, of most of these new visualizations— and mostly a sparing consumer at that, since economic and social history lagged behind political history as an area of research. Simple charts and graphs, such as pie charts,

line graphs, and histograms, were not difficult to interpret, and their visual conventions became part of what any ordinary reader would be expected to follow. As statistical analysis became more sophisticated, the visualizations that resulted became more and more central to the argument. In some cases, the visualization made interpretation possible. These success stories demonstrated the worth of statistical analysis and visualization. Perhaps the most notable example is John Snow's map of the incidence of cholera in an 1854 London outbreak, which helped plot the source of the outbreak at a single water pump in the neighborhood.4 Snow's cholera map showed that visualizations could serve as both narrative and analysis. Authors began to experiment with ways of using visual clues to tell complex stories about events, increasing the amount of information that could be conveyed in a small space and thereby overcoming the limitations of two dimensions in print.

19世纪社会科学的兴起，以及处理大型数据集能力的提高，催生了地图、时间线和谱系图之外的新型信息可视化方式。经过处理的数字信息，最适合用表格、图表和图形来表达。采用统计方法的数学、自然科学和社会科学走在图表发展前列。历史学在这些新型可视化面前主要是使用者，而不是设计者；而且使用得相当有限，因为作为研究领域的经济史和社会史本就落后于政治史。饼图、折线图、直方图等简单图表并不难理解，它们的视觉惯例逐渐成为普通读者也被期待掌握的阅读方式。随着统计分析越来越复杂，由此产生的可视化也越来越成为论证的核心。在某些情况下，正是可视化使解释成为可能。这些成功案例证明了统计分析和可视化的价值。最著名的例子也许是John Snow绘制的1854年伦敦霍乱暴发分布图，它帮助人们把疫情源头定位到社区中的一处水泵。4 Snow的霍乱地图表明，可视化既可以承担叙事功能，也可以承担分析功能。作者于是开始尝试利用视觉线索来讲述复杂事件，在有限空间中压缩更多信息，从而突破纸面二维展示的限制。

A noteworthy example of innovative presentation occurs in Charles Joseph Minard's 1869 Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812–1813, which portrays the advance and retreat of French troops in Russia on a scale map, showing the changing size of the force due to death and desertion through the thickness of the line representing the force.5 The conditions confronting the troops during the retreat are also illustrated by a time line of winter temperatures, graphically connected to the map-based chart. Though Minard was a civil engineer, not a historian, he was able to construct a very powerful single-page narrative of the events of Napoleon's march. Minard's chart

一个值得注意的创新呈现案例，是Charles Joseph Minard在1869年绘制的《1812—1813年法国军队俄国战役人员连续损失示意图》（Carte Figurative des pertes successives en hommes de l ' Arm é e Fran ç aise dans la campagne de Russie 1812 – 1813）。这张按比例绘制的地图呈现法国军队在俄国的推进和撤退，并通过代表军队的线条粗细变化，表现因死亡和逃亡而造成的兵力变化。5 撤退途中军队所面对的环境条件，则通过一条冬季气温时间线加以表现，并在图形上与地图连接起来。Minard虽然是一名土木工程师，而不是历史学家，却在一页纸上构造出了对拿破仑远征过程极具力量的叙事。Minard的图表

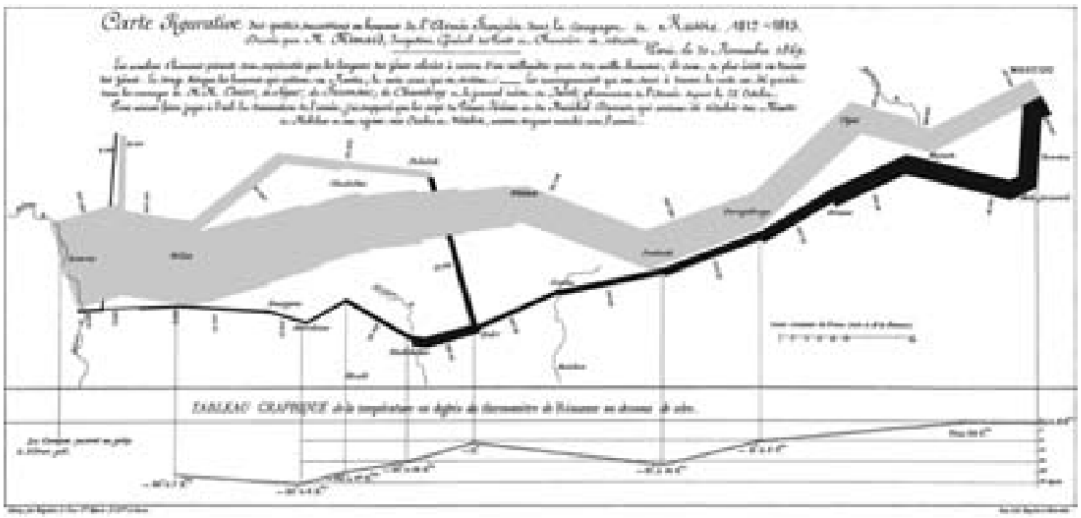


Fig. 7. Carte Figurative, by Charles Joseph Minard

图7：Charles Joseph Minard 的《Carte Figurative》

is often cited as a model example of information visualization because it is easy to understand, even for people with little background information on the topic or quantitative skills. The challenge for visualization is to be transparent, accurate, and rich in information. Minard's information-rich visualizations set the standard for both transparency and accuracy in the kind of work that could be done before computerization.

常被视为信息可视化的典范，因为即使读者几乎不了解这一主题、也缺乏定量分析技能，仍然很容易读懂它。可视化面临的挑战，是同时做到清晰透明、准确和信息丰富。在计算机化之前能够完成的可视化工作中，Minard这种信息高度丰富的图表为透明度和准确性都树立了标准。

As noted already, historians were mostly consumers of statistics-based visualizations from the social sciences, rather than innovators in constructing new kinds of visualizations. The advent of cliometrics and Annaliste total history in the 1960s forced more historians to become conversant with quantitative methods. Though the Annaliste approach to total history predated widespread use of the computer, much of the first wave of social scientific history relied on statistical packages like SPSS and SAS to process large amounts of data. The most determinedly quantitative works often had several pages of tables, most of which would be referenced in the text, though not always at the precise page that made the link between text and table most obvious. Instead of working as a driver of narrative, many of the tables and graphs produced in quantitative works of the 1960s, 1970s, and 1980s sat inert on the page, functioning more like the biographical pictures included in early historical works than as an integral part of the argument. Toggling between explication and evidence slowed reading considerably, so much so that readers of quantitative histories of the era sometimes broke into two broad groups: those who read the text and assumed the charts and graphs confirmed what was said there and those who read the charts and graphs while paying scant regard to the text. To be sure, many more historians developed the ability to rapidly interpret a greater variety of statistical representations. A researcher using a scatter plot with a line of best fit or a Lorenz curve comparing inequalities might reasonably expect that most readers would be persuaded by the results visible in the charts, without requiring significant textual explication. But most social histories continued to rely primarily on bar and line graphs as their most prominent visualizations. The fact that the statistical tools deployed often embedded assumptions that were inapplicable to the messiness of actual historical processes lent a false aura of scientific precision to very tentative conclusions. Many explanations have been offered for the relative decline of social history since its heyday in the 1970s. A failure of imagination in the integration of visualizations with text-based arguments may have contributed to the decline.6

如前所述，历史学家过去主要是社会科学统计可视化的使用者，而不是新型可视化的创新者。20世纪60年代计量史学和年鉴学派“总体史”的兴起，迫使更多历史学家熟悉定量方法。虽然年鉴学派的总体史理念早于计算机的广泛应用，但第一波社会科学取向的历史研究中，有相当一部分依赖SPSS、SAS等统计软件来处理大量数据。最强调定量分析的著作往往包含数页表格，正文会引用其中大多数表格，但表格并不总是恰好放在文字与数据联系最清楚的位置。20世纪60、70和80年代的许多定量历史著作中，表格和图形并没有推动叙事，而是静静地停留在页面上，更像早期历史著作中的传记人物插图，而不是论证不可分割的一部分。读者需要不断在解释文字和证据之间来回切换，显著拖慢阅读，以至于当时定量史的读者有时大致分成两类：一类只读文字，并默认图表印证了正文所说的内容；另一类则主要看图表，对正文几乎不加理会。诚然，越来越多历史学家已经能迅速理解更广泛的统计表达。一个使用带最佳拟合线的散点图，或用洛伦兹曲线比较不平等程度的研究者，完全可以合理预期多数读者会被图表中可见的结果说服，而不需要大量文字说明。然而，大多数社会史著作仍主要依赖柱状图和折线图作为最显著的可视化形式。所使用的统计工具往往内含一些并不适用于真实历史过程复杂性的假设，这给本来十分谨慎、暂时性的结论披上了一层虚假的科学精确性。对于社会史自20世纪70年代鼎盛之后相对衰落的原因，人们提出过许多解释；在将可视化与文字论证整合起来这件事上缺乏想象力，或许也是促成这种衰落的因素之一。6

While historians debated how to incorporate statistical methods into scholarship, statisticians were becoming more self-conscious about how the results of analysis were being used. This concern with how quantitative information was presented first came to the attention of most humanities scholars with the publications of Edward Tufte in the 1980s. His three key works—The Visual Display of Quantitative Information (1983), Envisioning Information (1990), and Visual Explanations

(1997)—placed the aesthetics and explanatory power of graphs and charts under closer scrutiny.7 Tufte was most responsible for renewing attention to Minard's Carte Figurative. Tufte's main target in his books was what he called "chartjunk," unnecessary clutter and contrived images that made visualizations confusing and sometimes deliberately misleading. Chartjunk was mostly associated with news and business publications in what today are called "infographics." Historians, like social scientists writing for scholarly publications, tended to avoid visual embellishments of charts and graphs, but they did have to be attentive to ensuring that legends were clear enough that readers were not deceived by how information was displayed. Even accurate information can mislead if it is presented in a way that creates false visual cues. For example, inexperienced readers might need some guidance with a logarithmic chart so that they do not mistake an exponential change for a linear one. Sometimes, visualizations can be shaped to seem more conclusive than the underlying data actually warrant. For example, if a line graph showing differences ranging from 65 percent to 85 percent has its baseline set at 50 percent rather than zero, it leads people to see the differences in values as starker than they actually are. These issues of what one might call rhetorical honesty in the formulation of visualizations were compounded for historians and other humanists by the hard choices that were required to generate the data to be processed in the first place. As Gibbs and Owens note in their contribution to this volume, historians have traditionally been told to mask the twists and turns of the research process in their finished work, to make their argument as strong as possible.8 This traditional approach can make any visualization seem like the product of a black box. Their proposal to share both data and methods in as transparent a manner as possible can have the additional benefit of making those visualizations easier to understand, because the logic of how and why they were generated is visible.

当历史学家还在争论如何把统计方法纳入学术研究时，统计学家却越来越自觉地反思分析结果被如何使用。20世纪80年代Edward Tufte的一系列著作，使大多数人文学者第一次真正注意到定量信息如何呈现这一问题。他的三部重要著作——《定量信息的视觉呈现》（The Visual Display of Quantitative Information，1983）、《设想信息》（Envisioning Information，1990）和《视觉解释》（Visual Explanations，1997）——把图表的审美和解释能力置于更严格的审视之下。7 Tufte在重新唤起人们对Minard《示意图》的关注方面尤其重要。他在书中主要批评自己所谓的“图表垃圾”（chartjunk），即那些没有必要的杂乱元素和刻意设计的图像，它们使可视化变得令人困惑，有时甚至具有故意误导性。“图表垃圾”主要出现在新闻和商业出版物中，也就是今天所说的“信息图”。历史学家和为学术出版物写作的社会科学家一样，通常会避免对图表进行装饰性加工，但仍必须确保图例足够清楚，不会因为信息的呈现方式而欺骗读者。即使信息本身准确，如果呈现方式制造出错误的视觉暗示，也可能造成误导。例如，缺乏经验的读者在阅读对数图时可能需要说明，否则会把指数变化误读为线性变化。有时，可视化的形式还会让结果看上去比数据真正能够支持的程度更加确定。例如，一张显示65%到85%差异的折线图，如果把基线设在50%而不是0%，就会让人觉得数值之间的差距远比实际更大。对于历史学家和其他人文学者而言，这种可以称为“可视化修辞中的诚实”问题，还会被另一个困难进一步放大：最初为了生成可处理的数据，本身就必须做出许多艰难选择。正如Gibbs和Owens在本书另一章中指出的那样，历史学家传统上被要求在最终成果中隐藏研究过程中的曲折和反复，以便让自己的论证显得尽可能有力。8 这种传统会使任何可视化看起来都像一个“黑箱”的产物。他们主张尽可能透明地共享数据和方法，这还会带来一个额外好处：读者能够看到可视化是如何、为何生成的，因此也更容易理解它。

Information-rich maps are a particularly good example of the challenge of balancing honesty in visual rhetoric and clarity and persuasiveness. One often has to come to a map visualization with sufficient background information to "read through" peculiarities of delivery. For example, maps of presidential elections that appear in almost any textbook of American history typically color in each state according to which candidate received the electoral votes. It is understood that the visual impact of the color contrast might over- or understate how close the vote actually was. Some states are large in area but small in population; others are the other way around. There are ways to make maps that account for those differences. For example, a cartogram adjusts the size of geographical areas to make them proportional to their populations, while a choropleth map uses shadings of color to indicate the strength of the victory in a given area.9 The two adjustments, cartogram and choropleth, can be combined to further the information density. Unfortunately, the distortions of the cartogram when combined with the choropleth can also make the information harder, rather than easier, to interpret, without a very high level of prior knowledge. This raises the question whether it is hard to

interpret such a visualization because the format of combining cartogram and choropleth is unfamiliar or because it makes unreasonable demands on the background information of the reader. Most historians have probably encountered a choropleth map and a cartogram in a print history book or contemporary source in the course of their research but probably have not encountered the two combined.

信息丰富的地图尤其能够说明这样一种挑战：如何在视觉修辞的诚实性，与表达的清晰度和说服力之间取得平衡。读者往往需要具备足够的背景知识，才能“穿透”地图呈现方式的特殊性来理解它。例如，几乎任何一本美国史教材中的总统选举地图，都会按照哪位候选人获得该州选举人票来给各州着色。人们知道，这种色彩对比的视觉冲击可能夸大或缩小实际选票差距。有些州面积很大但人口很少，另一些州则恰好相反。制图有办法对这种差异作出调整。例如，变形地图（cartogram）会调整地理区域的面积，使其与人口规模成比例；分级设色图（choropleth map）则用颜色深浅表示某一地区候选人的胜出幅度。9 这两种方式还可以结合使用，以进一步提高信息密度。但问题在于，变形地图造成的空间扭曲一旦与分级设色叠加，如果读者缺乏相当程度的先备知识，反而可能使信息更难、而不是更容易理解。这就提出一个问题：某种可视化之所以难懂，是因为“变形地图+分级设色”的形式还不够熟悉，还是因为它对读者的背景知识提出了不合理的要求？大多数历史学家在研究过程中可能都曾在印刷历史著作或当代材料中见过分级设色图和变形地图，但未必见过二者结合。

The question whether a visualization is hard to interpret because it is unfamiliar or because it relies on unrealistic expectations of background information assumes greater importance because digitization allows for even greater information density and novelty of form. Geospatial locating of information has been one of the richest areas of development in digital humanities. Complex visualizations based on maps are emerging as part of a “geospatial turn” in the humanities.10 One particular way that geospatial information density can increase is by animating it, adding time as another dimension of visualization. Just as a map can make one inch equal one mile, an animated time line can make one second equal one year. A simple combination of an animated map and a time line can create a powerful narrative without any text at all. A brilliant example of this is Isao Hashimoto’s animated map of the 2,053 nuclear explosions between 1945 and 1998, which dramatically narrates the contours of the nuclear age.11 Aside from the title, there is no background information associated with the animation. The only text in the piece is in the legend, which emerges as each new nuclear power first explodes a device. Sound, not words, is used as a second way of highlighting the data points. Yet, despite the absence of background information text, almost anyone watching the animation will come away with a deep understanding of the key features of the nuclear age. Only a modest background knowledge (such as knowing who the main antagonists in the Cold War were) makes the presentation of what might seem dry factoids not only informative but moving.

随着数字化使信息密度进一步提高、形式也更加新颖，“可视化究竟是因为形式陌生而难懂，还是因为它不切实际地要求读者具备过多背景知识”这一问题就变得更加重要。地理空间定位已经成为数字人文发展最丰富的领域之一，基于地图的复杂可视化正在人文学科的“地理空间转向”中不断出现。10 提高地理空间信息密度的一种方式，就是加入动画，把时间作为可视化的另一个维度。正如地图可以规定“一英寸等于一英里”，动画时间线也可以让“一秒钟等于一年”。一个简单的动画地图与时间线组合，甚至可以在完全没有文字的情况下形成强有力的叙事。Isao Hashimoto制作的动画地图就是一个出色例子：它呈现1945至1998年间的2,053次核爆炸，极具戏剧性地讲述了核时代的轮廓。11 除标题外，这段动画几乎没有任何背景说明；作品中唯一的文字位于图例中，并随着每一个新的核国家首次引爆核装置而出现。它还用声音而不是文字作为第二种强调数据点的方式。尽管缺少背景说明文字，几乎任何观看这段动画的人都能对核时代的关键特征形成深刻理解。只需具备有限的背景知识——例如知道冷战中的主要对立方是谁——那些看似枯燥的数据事实就不仅具有信息性，而且会产生情感力量。

Hashimoto’s animation of nuclear testing cannot be manipulated by the user, aside from pausing and resuming the animation. Edward Ayers has coined the term cinematic maps to describe map-based animations that show the process of change over time.12 The University of Richmond’s Digital Scholarship Lab has taken the traditional maps of presidential elections from 1840 to 2008 and turned them into an animated sequence.13 These maps try to overcome the information distortions caused by population differences and the electoral college, by providing not only county-level votes but a dot density map that shows the aggregated votes of five hundred voters. Instead of adjusting the size of the geographical area to make it fit the voting pattern, the dot pattern reflects the density

of votes in each region. Still, it is easy to imagine how this information could be converted to cartograms and choropleth maps of presidential elections to tell yet another story about changing voting patterns over the decades.

Hashimoto关于核试验的动画除了暂停和继续之外，用户无法进行其他操作。Edward Ayers提出“电影式地图”（cinematic maps）一词，用来描述通过地图动画展示随时间发生变化的过程。12 里士满大学数字学术实验室把1840至2008年的传统总统选举地图制作成了动画序列。13 这些地图试图克服人口差异和选举人团制度造成的信息失真：它们不仅提供县级选票数据，还使用点密度图，用一个点代表五百名选民的汇总选票。与其改变地理区域大小去迎合投票模式，不如让点的分布直接反映各地区的选票密度。当然，也很容易想象，这些信息还可以进一步转换成总统选举的变形地图和分级设色图，从而讲述几十年来投票模式变化的另一种故事。

As part of Stanford University's Spatial History Project, online visualizations have been created to accompany Richard White's recent book on the development of the transcontinental railroad.14 This project is particularly interesting for understanding the impact of digital humanities on current historical practice, because it is directly associated with a print work and seems likely to serve as a template for future hybrid productions of print and digital. It is also closely aligned with a still more expansive set of visualizations from the Stanford Spatial History Project that relate to themes of the book, collected under the heading Shaping the West.15

作为斯坦福大学“空间历史项目”（Spatial History Project）的一部分，研究者制作了在线可视化来配合Richard White最近关于横贯大陆铁路发展的著作。14 这个项目对于理解数字人文如何影响当下历史研究实践尤其有意义，因为它直接与一本纸质著作相联系，而且很可能成为未来“纸质出版+数字成果”混合型出版的模板。它还与斯坦福空间历史项目中一组规模更大的可视化密切相连，这些可视化围绕该书主题展开，并被汇集在“塑造西部”（Shaping the West）这一标题之下。15

There are twenty-six different visualizations included at the site, sixteen of which are animated. Not unexpectedly, several of the animations simply plot space and time, like the animations previously described. But others complicate the visualization by layering information in innovative ways. For example, one visualization reframes shipping distances in California in terms of not just track length but also time to delivery and cost. Even though the animations have been created as an accompaniment to an academic work, they offer an interactive opportunity to the reader that those other animations do not. Readers can customize the presentation of data to isolate issues of particular interest to them, rather than depending on the author to frame the question being answered. Interactive engagement with a visualization is yet another innovation made possible by digitization.

该网站一共包含26种不同的可视化，其中16种是动画。不出所料，一些动画只是像前面提到的例子那样，把空间和时间绘制出来；但另一些则通过创新的信息叠加方式，使可视化变得更加复杂。例如，有一项可视化重新解释加利福尼亚的运输距离，不仅考虑铁路线长度，还同时纳入交付时间和成本。尽管这些动画是作为学术著作的配套内容制作的，它们却为读者提供了前述动画所不具备的互动机会。读者可以自定义数据的呈现方式，把自己特别关心的问题单独提取出来，而不必完全依赖作者来规定“正在回答什么问题”。这种与可视化的互动式接触，也是数字化带来的另一项创新。

As historians ask more complex questions about the data they have assembled, the problem of how best to present the information requires more thought. On the Shaping the West site, each visualization has an “About” or “Help” tab that functions as a legend and guide to the information contained in the site. The visualizations are not self-explanatory. A particularly complex visualization links the geography of the railroads with a network diagram of the boards of directors and sources of capital for each. The “About” tab for that visualization includes a “How to Read” statement for the graphic. Such “How to Read” statements recognize that the visual vocabulary of innovative sites may not be familiar enough to make an argument without further explication of methods.

随着历史学家对自己汇集的数据提出更复杂的问题，如何以最佳方式呈现信息也需要更认真地思考。在“塑造西部”网站中，每一种可视化都有“About（关于）”或“Help（帮助）”标签页，承担图例和信息阅读指南的功能。这些可视化并不是不言自明的。其中一项特别复杂的可视化，把铁路的地理分布与各铁路公司董事会和资本来

源的网络图连接起来；该图的“ About ”页专门提供了一段“ How to Read（如何阅读） ”说明。这类“ 如何阅读 ”的说明承认：创新网站所使用的视觉词汇还没有熟悉到可以在没有方法解释的情况下独立完成论证。

Undoubtedly the biggest advocate for the rhetorical power of statistical animations that incorporate interactive features is the Swedish statistician Hans Rosling. Using a tool called Gapminder, he has created an animation of life expectancy at birth and per capita GDP since 1800 for all countries, to demonstrate the evolution of world health.16 One can “play” Rosling’s animation in a noninteractive mode to see the story he tells. Color coding differentiates countries in different parts of the world. If one scrolls over the circles on the chart, one can see which country each represents. Circles vary in size depending on the population of the country and on change in size over time in response to population growth, so the reasonably well-informed viewer can quickly locate major countries like China, India, and the United States even without scrolling over the circles. The animation contains deep layering of information that is easy to interpret, even without an extensive background. In videos where he talks about the data, Rosling shows that the information illustrates a dramatic narrative of the convergence of the world on higher levels of health and wealth, but the point comes across perfectly clearly even without verbal accompaniment.17

毫无疑问，最积极主张带有互动功能的统计动画具有修辞力量的人之一，是瑞典统计学家Hans Rosling。他利用Gapminder工具制作了一套动画，展示自1800年以来世界各国的出生预期寿命和人均GDP，用来说明全球健康状况的演变。16 人们可以在非互动模式下“ 播放 ”Rosling的动画，观看他所讲述的故事。颜色编码用来区分世界不同地区的国家；把鼠标移到图中的圆点上，就能看到圆点代表哪个国家。圆的大小取决于该国人口规模，并且会随着人口增长而随时间变化，因此一个具备一般知识的观看者，即使不把鼠标移动到圆上，也能很快识别中国、印度和美国等大国。动画把多层信息叠加在一起，却仍然很容易理解，并不要求读者具备很深的背景知识。在介绍这些数据的视频中，Rosling说明这些信息讲述了一个极具戏剧性的故事：世界逐渐趋向更高的健康和财富水平；但即使完全没有口头讲解，这一观点本身也表达得非常清楚。17

Rosling draws on an extremely rich database, and readers are able to customize the display of information according to their own interests. One tab allows readers to orient the circles on a map of the world rather than on two axes of a chart. Another allows readers to choose which country’s data to include or not include in the animation. If one wants to isolate countries from a single continent or countries that start out a similar size, one can do so. One can also adjust the time line to focus on narrower periods where crucial changes might be taking place, instead of having to go through the entire time span for “big picture” changes. Because of this option for customization, Rosling’s project both makes an argument that is explicit in the first animation one sees and provides the basis for a reader’s further exploration of his or her own interests. While the kind of graphs used by Rosling were developed prior to the web, they posed real challenges of presentation and interpretation in the two-dimensional format of print. Animation increases their interpretive force dramatically.

Rosling使用的是一个极其丰富的数据库，读者可以根据自己的兴趣自定义信息的显示方式。一个标签页允许读者把圆点放到世界地图上，而不是放在图表的两个坐标轴之间；另一个标签页允许读者选择哪些国家的数据要纳入动画、哪些不纳入。如果想只观察某一洲的国家，或者一开始规模相近的国家，都可以做到。读者还可以调整时间线，把注意力集中到可能发生关键变化的较窄时间段，而不必为了理解“ 整体趋势 ”而从头看完整个时间跨度。正因为存在这种自定义选项，Rosling的项目一方面在读者看到的第一个动画中提出明确论点，另一方面又为读者依据个人兴趣继续探索提供了基础。Rosling使用的这类图表虽然早在网络出现之前就已发展出来，但在二维印刷形式中会面临很大的呈现和解释困难。动画显著增强了它们的解释力量。

Animation and reader control of the data stream are not the only ways in which digitization affects interpretation. Websites are new enough that there are still opportunities to subvert standard expectations and make readers more attentive to how visual cues structure an argument (in ways that are much less costly than trying to subvert visual cues in print media). A good example is Whitney Trettien’s “Computers, Cut-Ups, and Combinatory Volvelles: An Archaeology of Text-Generating Mechanisms,”18 which encourages a nonlinear reading of her argument about nonlinear texts. The front page of the site does not offer a table of contents or an obvious sequential path through the material. It has a grid of fourteen by twenty-one white squares that light up and change color either when scrolled over or when specific pages of text are clicked. Color coding allows

readers to see which sections of the website deal with specific themes, creating a second way of envisioning the argument. Physical proximity of squares and color groupings work together to create a structure to the argument that is as easy to see as it is to read. Interestingly, the idea of color coding in a grid format was presaged in Elizabeth Peabody's nineteenth-century Universal History: Arranged to Illustrate Bem's Charts of Chronology.19 Trettien's experiment with a visual "breadcrumb trail" makes it possible to reimagine how arguments can be presented in an environment where the reader controls what page to turn to next.

动画以及读者对数据流的控制，并不是数字化影响解释的唯一方式。网站仍然是一种相对新的媒介，因此还有机会打破标准化预期，使读者更加注意视觉线索如何组织论证；而这样做的成本，也远低于在印刷媒介中颠覆既有视觉惯例。一个很好的例子是Whitney Trettien的《计算机、剪贴与组合式转盘：文本生成机制的考古学》（“Computers, Cut-Ups, and Combinatory Volvelles: An Archaeology of Text-Generating Mechanisms”）。18 这一项目鼓励读者以非线性的方式阅读她关于非线性文本的论证。网站首页没有目录，也没有一个明显的顺序路径；页面上是一张由14×21个白色方格组成的网格，当鼠标滑过或点击某些文字页面时，方格会亮起并改变颜色。颜色编码让读者看出网站的哪些部分处理哪些主题，于是形成了理解论证的第二种方式。方格在空间上的接近程度和颜色分组共同构成论证结构，使它既可以被“看见”，也可以被“阅读”。有意思的是，19世纪Elizabeth Peabody的《世界史：配合Bem年代图表编排》（Universal History: Arranged to Illustrate Bem’s Charts of Chronology）已经预示了这种网格中的颜色编码理念。19 Trettien设计的视觉“面包屑路径”实验，使我们可以重新想象：在读者自己控制下一页要去哪里时，论证还可以如何被呈现。

As Peabody's work shows, it is possible to use color as a visual cue in print texts, but it is generally prohibitively expensive. Online, color is both efficient and cost-free. A superb example of using color to highlight relationships in text is Ben Fry's concordance of the six editions of Charles Darwin's Origin of Species completed in his lifetime.20 The original text is represented by each sentence being compressed into a single line. The reader can scroll over each line to get a text box of the sentence. The additions in subsequent editions are represented by different colored lines. The colors allow one to quickly grasp, for example, that chapter 4 was most extensively revised in the third and fifth editions, while chapter 6 was most extensively revised in the fourth edition. Dramatic changes are visible because of tiny lines of color. This same principle is taken up in the WordSeer project at the University of California, Berkeley. The WordSeer researchers have digitized a corpus of printed slave narratives and compressed each narrative to a single bar of a heat map. One can search for words across all of the narratives and see how frequently they turn up in each of the paragraphs, represented by lines within each bar.21 Color, brightness, and lines and bars thus become powerful ways of making interpretive leaps about texts, as long as one has the background knowledge to understand the implications of the visualization.

Peabody的作品表明，在印刷文本中也可以用颜色作为视觉线索，但通常成本高得难以承受。在网络环境中，颜色既高效又几乎没有额外成本。利用颜色突出文本关系的一个优秀案例，是Ben Fry对Darwin生前完成的六个《物种起源》版本所做的对勘可视化。20 原始文本中的每一句都被压缩成一条线；读者把鼠标移到某一条线上，就会出现包含该句文字的文本框。后续版本中新增的内容则由不同颜色的线表示。通过这些颜色，人们可以迅速看出，例如第四章在第三版和第五版中的修改最多，而第六章在第四版中的修改最多。很小的彩色线条就足以让重大变化显现出来。加州大学伯克利分校的WordSeer项目也采用了类似原则。研究人员把一批印刷出版的奴隶叙事数字化，并把每一篇叙事压缩成热力图中的一根长条。用户可以在全部叙事中检索词语，并看到它们在每个段落中出现的频率，这些频率由长条内部的线条表示。21 因而，只要读者具备理解可视化含义所需要的背景知识，颜色、亮度、线条和条带都可以成为对文本进行解释性跨越的强大工具。

Much historical writing is implicitly or explicitly about network connections, but historians are less familiar with how social scientists have been visualizing networks than they are with standard statistical visualizations. The Mapping the Republic of Letters project at Stanford University shows how network visualizations can be used in historical work.22 It overlays a networked map of correspondents on the actual map of Europe, with each link in the visualization representing a letter sent between an author at one location and a reader at another. The network described at Mapping the Republic of Letters is personal rather than conceptual, as was the network of railroad boards of

directors previously mentioned. The railroad visualization was more complicated, because the nodes of the network were not linked to a map but were a pure visualization of relationships. The online prototype of visualizations of network relationships is Thinkmap's Visual Thesaurus, which allows readers to move from node to node in pursuit of related concepts.23 Visualizing networks poses several dangers for historians. First of all, a network theory of graphs adheres to mathematical principles that have little relation to lived human experiences. In an effective network visualization, the location of nodes is not predetermined but is specified by the nature of the links between them. If you remove one source of links from the analysis, the location of nodes may become different. As one undertakes a more complicated kind of network analysis in the mode of Visual Thesaurus, it is more likely that following any single link trail can quickly get one lost in the thicket of concepts. Unless one understands the algorithms being used to create the nodes, it is extraordinarily difficult to understand why nodes are in a specific relationship to one another. Thus network analysis demands the kind of "hermeneutics of data" advocated by Gibbs and Owens in this volume. But even when the concepts and relationships being illustrated are relatively straightforward, the task of visualizing them can prove complicated by the volume of connections being analyzed. The sheer density of nodes can make it hard to single out factors that might interest the reader. For example, in the network visualization of those scholars who make up the "vizosphere" (the leading edge of discussion about the future of visualizations), there is a barely differentiated blob of circles instead of a clear pattern of lines between sites.24

大量历史写作都或明或暗地涉及网络联系，但与标准统计可视化相比，历史学家对社会科学家如何可视化网络要陌生得多。斯坦福大学“书信共和国地图”（Mapping the Republic of Letters）项目展示了网络可视化如何用于历史研究。22 它把通信者网络叠加到真实的欧洲地图上，可视化中的每一条连线都代表一封从某地作者寄往另一地读者的信。“书信共和国地图”所描绘的网络是人与人之间的网络，而不是概念网络，这一点与前面提到的铁路公司董事会网络相似。不过，铁路可视化更加复杂，因为其网络节点并不与地图位置绑定，而是一种纯粹的关系可视化。网络关系可视化的一个在线原型，是Thinkmap的Visual Thesaurus，它允许读者从一个节点移动到另一个节点，追踪相关概念。23 对历史学家而言，网络可视化也存在若干危险。首先，图论意义上的网络遵循数学原则，而这些原则与人类真实生活经验之间并没有直接对应关系。在一个有效的网络可视化中，节点位置不是预先规定的，而是由节点之间连接的性质决定。如果从分析中移除一种连接来源，节点的位置就可能发生变化。当网络分析像Visual Thesaurus那样变得更加复杂时，沿着任何一条连接路径追踪，都很可能很快让读者迷失在概念丛林中。如果不了解生成这些节点的算法，就极难理解为什么节点之间会呈现某种特定关系。因此，网络分析需要Gibbs和Owens在本书中所提倡的那种“数据诠释学”。不过，即使要展示的概念和关系本身比较简单，连接数量过多仍会使可视化变得复杂。节点密度太高，会让读者很难找出自己关心的因素。例如，一张把“vizosphere”（围绕可视化未来的前沿讨论圈）中的学者可视化出来的网络图，最终呈现的几乎只是一个难以区分的圆点团，而不是各网站之间清晰可见的连线结构。24

Innovative visualizations have entered the mainstream of online user experience in the professions and social sciences. Just as SPSS, SAS, and, later, R were created to enable basic statistical analysis, programs like Gephi have been created to undertake network analysis. In the wake of Tufte's work, numerous authors now write about information design, though again mostly targeting a business and journalism audience. Every day, sites like FlowingData highlight innovative uses of visualization to make new arguments, such as PeopleMovin, which illustrates migration flows between countries.25 It is clear from these sites that people are still expanding the realm of the possible in visualizing information. Looking over these visualizations, even when they are not explicitly historical, will give historians strategies for making more powerful arguments to complement and sometimes even substitute for text. But the task of building those arguments will have to include educating fellow historians about how to interpret visualizations. As we have already noted, it can be very difficult for the uninitiated historian to intuit relationships between entities in a network analysis when they are put into a visualization scheme. Yet networks are often at the center of questions of greatest interest to historians. To the extent that the difficulties in interpreting innovative visualizations like interactive network diagrams are caused by a simple lack of familiarity with them, they can be overcome by building more such sites. To the extent that they are caused by a lack of background knowledge to understand the cues, creators of such sites will have to learn to build new ways of incorporating that background information as economically, in the use of text, as possible. In either

case, at some point, historians will have to accustom themselves to “reading” network diagrams as adeptly as they read maps or scatter plots.

创新型可视化已经进入专业领域和社会科学的主流在线用户体验。正如SPSS、SAS以及后来R的出现，是为了让基本统计分析更容易进行，Gephi之类的软件则是为了进行网络分析而开发。Tufte的著作之后，越来越多作者开始讨论信息设计，尽管这些讨论的主要受众仍然是商业和新闻行业。FlowingData等网站每天都会展示利用可视化提出新论证的创新案例，例如展示国家间人口迁移流动的PeopleMovin。25 从这些网站可以看出，人们仍在不断扩展“信息究竟可以怎样被可视化”的可能范围。即使这些可视化并非直接讨论历史问题，历史学家也可以从中学习如何构造更有力量的论证，用来补充文字，有时甚至取代文字。不过，要建立这种论证，也必须教会其他历史学家如何解释可视化。正如前面已经指出的，对于没有接受过相关训练的历史学家来说，当网络分析中的实体关系被转换成可视化方案时，很难凭直觉理解这些关系；然而，网络偏偏又经常处在历史学家最关心的问题中心。如果互动网络图等创新可视化之所以难懂，只是因为人们还不熟悉它们，那么制作更多这样的站点就能逐渐克服这一问题。如果困难来自读者缺少理解视觉线索所需的背景知识，那么站点设计者就必须学习如何尽可能节省文字，同时把这些背景信息纳入其中。不论是哪一种情况，历史学家最终都必须学会像阅读地图或散点图一样熟练地“阅读”网络图。

Notes

注释

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6. One can overstate the degree of that decline. Social history was always something of a grab-bag term that included nonquantitative approaches. Nevertheless, the relative decline of social history in comparison to cultural history is unmistakable. See Robert B. Townsend, "Decline of the West or Rise of the Rest?," Perspectives on History, September 2011, http://www.historians.org/perspectives/issues/2011/1109/1109pro1.cfm.

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